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Deep Learning-Based Hardware Trojan Detection via Dual-Path CNN and Markov Transition Field Representations

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This paper introduces a dual-path deep learning framework specially designed for the efficient detection and classification of Hardware Trojans (HTs) through Side-Channel Analysis (SCA). Using Markov Transition Field (MTF) encoding and a reshaping strategy, the proposed method first converts side-channel time-series signals, including power traces, electromagnetic leaks, and timing data, into two different image-like formats. Detecting subtle Trojan activity requires these representations to capture intricate signal dynamics. Subsequently, each image is processed by a separate convolutional neural network (CNN) branch within the dual-path architecture, with each path optimized for extracting complementary features. To enhance classification performance, the outputs of both CNNs are fused via a dense neural network layer. The dual-path mechanism contributes to improved detection accuracy and robust feature extraction, while the overall architecture supports the classification of specific Trojan types. Evaluation using the publicly available AES Hardware Trojan dataset sourced from TrustHub and IEEE DataPort demonstrates that the proposed model benefits from CNN-based feature learning and outperforms several existing approaches.

Keywords: Hardware Trojan Detection, Time-Series Analysis, Markov Transition Field, Convolutional Neural Networks, Deep Neural Networks, Side-Channel Analysis.

1. Introduction

The security and integrity of electronic systems are paramount in today's increasingly interconnected and technology-dependent world. Among the most serious threats to these systems are Hardware Trojans (HTs), malicious modifications inserted into electronic components during the design, manufacturing or supply chain phases¹. Operating at the hardware level, HTs are significantly more difficult to detect and neutralize than software-based attacks due to their deep integration and limited observability within hardware operations^{2,3}.

The globalization and complexity of the semiconductor supply chain further exacerbate this risk, offering adversaries more opportunities to introduce subtle, yet impactful, alterations. In critical domains such as national security, healthcare,

and finance, such Trojans can disrupt essential functions, leak sensitive data, and trigger severe consequences ^{4,5}. Due to their stealthy nature and limitations in conventional testing methods, HT attacks often go undetected, making them a major concern for cybersecurity professionals. This growing threat underscores the urgent need for robust and intelligent detection mechanisms ^{6,7}.

One promising solution is Side-Channel Analysis (SCA), which monitors physical parameters such as power consumption, electromagnetic emissions and timing characteristics to detect anomalies that could indicate malicious hardware modifications. Despite their effectiveness, conventional SCA techniques are often affected by environmental noise, device-level variations, and the lack of a trusted reference, leading to potentially false positive or negative results ⁸. To overcome these limitations, machine learning (ML) has been increasingly integrated with SCA to improve detection capabilities ⁹. In these hybrid approaches, ML algorithms are trained to learn the normal physical and operational behavior of hardware components. This enables the automatic identification and classification of subtle anomalies in side-channel signals, many of which may indicate the presence of HTs.

Building on these advances, various machine learning techniques, both supervised and unsupervised, have been successfully applied to detect HT attacks by analyzing side-channel signals ^{10,11,12}. Supervised methods such as Support Vector Machine (SVM) ¹³, Naïve Bayes ^{14,15}, natural language processing (NLP) ¹⁶, Random Forest ¹⁷, and XGBoost ¹⁸ have been shown to be effective in identifying anomalies indicative of HTs. The recent work of ¹⁹ explores a machine learning-assisted channel analysis on the power supply side and improves classification accuracy on different Trojan states. Another approach proposed in ²⁰ utilizes ensemble learning that combines multiple weak classifiers to efficiently detect HTs and achieves high performance across multiple TrustHub datasets. In addition, innovative techniques such as Deep Belief Networks (DBN) optimized with Stacked Autoencoder Hashing Optimization ²¹ and Deep Transfer Learning ²² represent state-of-the-art approaches that improve detection accuracy and adaptability in the field of HT attack identification. Deep learning architectures, especially CNNs, are widely used for HT detection due to their ability to extract complex patterns from side-channel signals. In ²³, an attention mechanism was integrated into CNNs feature extraction. Recent research demonstrates the effectiveness of ResNeXt networks in detecting Trojans ²⁴ by using hierarchical feature learning for multi-Trojan detection scenarios. The combination of SCA and CNN improves feature extraction and provides robustness to noise, making it an effective tool for HT detection. By converting time-series signals into image-based representations, CNNs can efficiently handle high-dimensional data, reducing the need for manual feature extraction and providing a scalable solution for modern cybersecurity challenges. ^{25,26}.

In contrast to 1D CNNs that process time series data directly, 2D CNNs need to transform these data into image-like formats to fully exploit their spatial pattern recognition capabilities. For this transformation, Markov Transition Fields (MTFs) can be used, which convert time series signals to matrix representations that en-

code transition probabilities between states²⁷. This representation preserves the temporal dynamics of the signal and explicitly captures the relationships between states, which is crucial for accurate HT detection. As a result of using MTFs, CNNs can detect subtle anomalies and hidden dependencies that are commonly hard to detect when viewing raw time series data. Through the integration of MTFs into CNN architectures, it becomes possible to discover complex patterns indicative of Trojan activity, thus strengthening hardware security in adversarial environments. Unlike prior work that uses single-path or sequence-based deep learning models, our method introduces a dual-path CNN architecture that fuses two distinct 2D image representations—Markov Transition Field (MTF) and reshaped time-series matrices—to capture both fine-grained transition dynamics and coarse-grained structural patterns from side-channel signals.

Recent studies using time-series-to-image encodings address markedly different problems. ²⁸ proposed a multi-representational fusion scheme for remote sensing classification by applying various image encodings (MTF, GAF, recurrence plots) to vegetation time-series data. While MTF is a shared technique, their goal was spatial land-cover classification, and their architecture combined encodings via ensemble classifiers. In contrast, our method directly fuses MTF and reshaped CNN feature representations for Trojan detection in cryptographic hardware. ²⁹ investigated multiple time-series-to-image transformations (including MTF) for unsupervised anomaly detection in aeronautical vibration monitoring. Their study emphasizes encoding robustness and single-path CNN performance, not multi-path fusion or Trojan classification. Our work differs by using two complementary encodings and applying them to side-channel data from AES-based devices with known Trojan types. ³⁰ introduced a CNN-Attention-LSTM model trained on netlists from the TrustHub benchmark. While effective in detecting complex Trojan-induced anomalies in circuit logic, their approach relies on processing raw netlist-derived features as sequential data. In contrast, our method differs fundamentally in three aspects. it uses power traces rather than netlist features as input, transforms the data into image-based representations instead of sequences, and performs both Trojan detection and type classification rather than detection alone.

This paper introduces a single-stage deep learning architecture for detecting and identifying HTs using SCA and time-series imaging. Building upon our preliminary work presented at ICM 2024 ²⁶, the proposed framework transforms raw time-series power traces into two complementary image-like formats: (i) reshaped 2D grids, which preserve the global statistical structure and allow CNNs to extract coarse-grained spatial correlations, and (ii) MTF representations, which encode the temporal transition probabilities between quantized signal states, capturing fine-grained temporal dependencies and dynamic patterns. These representations are processed in parallel through two dedicated CNN pathways within a dual-path architecture, allowing the model to take advantage of both structural and dynamic characteristics of the side-channel signal. The integrated feature maps are then

fused for joint classification, allowing the network to detect the presence of an HT and identify its specific type.

In addition, an extensive experimental evaluation is performed using publicly available AES Hardware Trojan datasets sourced from TrustHub and IEEE DataPort. The paper conducts both binary (normal vs. specific Trojan type) and multi-class (multiple Trojan types simultaneously) classification tasks, analyzing class-specific performance through metrics such as confusion matrices, Receiver Operating Characteristic (ROC) curves, and Precision-Recall (PR) curves. Hyperparameter optimization is also systematically explored to determine the most effective configuration of convolutional filters and kernel sizes, enhancing the model's generalization and accuracy. Finally, latent space visualizations using t-distributed stochastic neighbor embedding (t-SNE) are employed to provide insights into the model's feature extraction capabilities and identify challenging cases prone to misclassification.

The remainder of this paper is structured as follows. Section 2 provides a foundational background on Hardware Trojan threats and reviews prominent detection strategies. Section 3 introduces the proposed approach, emphasizing the conversion of time-series signals into image-based formats. In Section 4, we detail the experimental evaluation and discuss the results obtained from our framework. Finally, Section 5 offers concluding remarks and highlights key outcomes of the paper.

2. Overview of Hardware Trojans

It is important to understand HT attacks as they pose a serious threat to the security and reliability of electronic systems, including critical infrastructure and consumer electronics, and lead to serious risks that can result in catastrophic failures and privacy breaches.

HTs can be categorized based on their activation triggers and malicious payloads³¹. The trigger mechanism refers to the specific condition or set of conditions that activate the HT. These triggers are often designed to remain hidden and undetected until a specific event occurs. They can range from simple events, such as a specific date or time, to complex patterns of activity and specific inputs that only occur in rare circumstances. This advanced triggering mechanism allows the Trojan to evade detection during standard testing and normal operation, ensuring that it remains inactive until it can fulfill its intended purpose. Once the HT is activated, the payload represents the malicious action that the Trojan performs. Depending on the payload, a Hardware Trojan can alter device functionality, spy on confidential information, affect system performance or cause physical damage. It is possible to insert HTs at various stages of a chip's lifecycle, from the initial design phase to fabrication and deployment. During the design phase, third-party intellectual property (IP) cores, often used to reduce development time and cost, can be compromised by surreptitiously introducing HTs. Design software tools are also vulnerable, as malicious code or circuitry can be embedded, remaining undetected

until triggered by specific conditions. The fabrication stage, particularly when outsourced to untrusted facilities, is another critical vulnerability point where chip layouts can be altered or lithography masks manipulated to embed HTs directly into silicon. Although global outsourcing is economically advantageous, the complexity and microscopic scale of modern integrated circuits make such alterations nearly undetectable. Additional vulnerabilities exist during the assembly and packaging stages, where HTs can be inserted into the chip package by modifying the silicon or embedding malicious components. Chips are also exposed to the risk of interception and tampering as they move through the supply chain, representing another layer of vulnerability.

Given the complexity of these threats, robust security measures must be implemented throughout a chip's lifecycle. A holistic approach is essential to mitigate the risks posed by HTs and ensure the security and functionality of electronic systems. This approach should include secure and verified design practices, integrity checks during manufacturing and assembly, and supply chain security. In addition, the development and deployment of advanced detection techniques is crucial to identify and neutralize HTs that evade initial preventive measures. Without effective HT detection mechanisms, critical electronic infrastructure remains vulnerable to significant compromise.

2.1. Threat Model

In this work, we assume an adversarial setting in which the attacker is capable of inserting HTs during the design or fabrication stages. This may occur through compromised third-party IPs or untrusted foundries. The HTs are assumed to be stealthy and dormant during regular testing procedures, becoming active only under rare or specific input conditions. This aligns with common trigger mechanisms discussed earlier, such as specific patterns or rare event activation. We assume that the detection must rely on side-channel measurements, specifically power traces, collected during normal device operation. These measurements are assumed to be acquired in a trusted environment using standard equipment. The goal of the defender is to detect the presence of HTs and classify their type based solely on observed deviations in the side-channel signal behavior.

3. Dual-Path Deep Learning Framework for Hardware Trojan Detection

This section describes the proposed methodology for detecting and identifying HT threats. Time series data are converted into two 2D images using MTF and reshaping techniques so that CNNs can apply their spatial feature learning capabilities. The extracted features are then classified to detect the presence of an HT attack and identify the specific Trojan type.

3.1. Time-Series-to-Image Transformations for Side-Channel Data

To enable the use of CNNs for analyzing side-channel signals, this paper transforms raw time-series power traces into image-like representations that retain both structural and temporal information essential for accurate HT detection. The transformation of time series data into image-like representations has led to significant improvements in the ability of deep learning models—particularly CNN—to extract spatial and temporal patterns from complex sequential data. This approach has proven effective across a wide range of domains, including speech recognition, physiological signal classification, financial forecasting^{32,33,34}.

The dual-path architecture proposed in this paper uses both reshaped and MTF-based representations to enhance feature learning from side-channel power traces. The reshaped format enables the model to capture coarse-grained structural patterns by organizing the time-series data into a fixed 2D matrix layout, preserving the sequential order of signal values. In contrast, the MTF representation enriches the feature space by encoding temporal transitions and state-level dependencies into a spatial format that is particularly well-suited for convolutional analysis. This section provides a detailed explanation of the two transformation techniques employed in the proposed dual-path framework. The first subsection focuses on the reshaping method, which highlights structural regularities within the signal, while the second subsection describes the MTF technique, which captures fine-grained temporal dynamics. Together, these representations serve as complementary inputs to parallel CNN pathways, enabling more comprehensive and robust feature extraction for accurate Hardware Trojan detection.

3.1.1. Feature Representation via Reshaping

One straightforward approach to converting one-dimensional time-series data into a two-dimensional format is the *reshaping technique*, where the linear signal is reorganized into a matrix by arranging data points in row-major order³⁵. Given a time-series sequence $\mathbf{x} = \{x_1, x_2, \dots, x_n\} \in \mathbb{R}^n$, this method partitions $\mathbf{x}$ into a matrix $\mathbf{X} \in \mathbb{R}^{r \times c}$ such that $n = r \cdot c$, where r and c represent the number of rows and columns, respectively.

The reshaping process is defined as:

$$\mathbf{X}_{i,j} = x_{(i-1) \cdot c + j}, \quad \text{for } 1 \leq i \leq r, 1 \leq j \leq c$$

This preserves the sequential order of the data while mapping it into a spatial structure that can be efficiently processed by 2D convolutional layers. Unlike transformations that alter the signal's representation (e.g., spectrograms or MTF), reshaping retains the raw amplitude information and introduces a spatial locality that CNNs can exploit.

Although reshaping does not explicitly encode temporal dependencies, it provides a coarse structural representation that can reveal macro-level variations in sig-

nal shape, energy distribution, or repetitive patterns—especially valuable in SCA. In the context of Hardware Trojan detection, such patterns can arise from subtle changes in circuit activity when Trojans are activated. By processing these reshaped matrices through a dedicated CNN path, the model is able to learn discriminative features associated with normal and compromised circuit behavior.

3.1.2. Feature Representation via MTF

Another innovative technique is the MTF representation, where time series data is transformed into a matrix that represents transition probabilities and captures the underlying dynamics and dependencies between different states in the data³⁶. This matrix illustrates the probability of transitions between different quantized states and provides a detailed visual representation of the temporal dynamics within the time series³⁷. Conversion of time series data into a transition probability matrix enables the visualization and analysis of complex temporal patterns in two-dimensional space. The MTF technique is particularly advantageous because it encodes both temporal variations and state dependencies in spatial patterns, making it ideal for use with two-dimensional CNNs. These CNNs are extremely effective at identifying spatial relationships and patterns within the data, enhancing the model's ability to recognize hidden features and intricate patterns³⁸.

First, consider a normalized time-series signal represented as a sequence of real values:

$$\mathbf{x} = \{x_1, x_2, \dots, x_n\}, \quad x_t \in \mathbb{R}$$

The data is normalized to the range $[-1, 1]$ to standardize its scale and improve numerical stability:

$$\tilde{x}_t = \frac{2(x_t - \min(\mathbf{x}))}{\max(\mathbf{x}) - \min(\mathbf{x})} - 1 \quad (1)$$

Next, the normalized sequence is quantized into k discrete levels. Each value $\tilde{x}_t$ is mapped to a bin between 0 and $k - 1$:

$$\hat{x}_t = \left\lfloor k \cdot \frac{\tilde{x}_t - \min(\tilde{\mathbf{x}})}{\max(\tilde{\mathbf{x}}) - \min(\tilde{\mathbf{x}})} \right\rfloor \quad (2)$$

Here, $\hat{x}_t \in \{0, 1, \dots, k - 1\}$ denotes the quantized state at time step t , and k is the number of quantization bins. The transition probabilities between quantized states are then computed. For each state pair (i, j) , the Markov transition probability from state i to state j is given by:

$$P_{ij} = \frac{\sum_{t=1}^{n-1} \mathbb{1}(\hat{x}_t = i \wedge \hat{x}_{t+1} = j)}{\sum_{t=1}^{n-1} \mathbb{1}(\hat{x}_t = i)} \quad (3)$$

where $\mathbb{1}(\cdot)$ is the indicator function that evaluates to 1 if the condition is true, and 0 otherwise. The final MTF is a $k \times k$ matrix defined as:

$$M_{ij} = P_{ij} \quad (4)$$

Each entry M_{ij} represents the likelihood of transitioning from state i to state j , forming a two-dimensional representation that encodes the temporal transition dynamics of the original time series. This matrix can be visualized as a grayscale image where pixel intensities correspond to transition probabilities. By transforming the time series into an MTF matrix, we convert temporal dependencies into a spatially structured format, enabling the application of 2D CNNs for downstream tasks such as classification, anomaly detection, or behavioral analysis in side-channel contexts like Hardware Trojan detection.

It is important to note that the resolution of the resulting MTF image can be adjusted by modifying the time-series length or the number of quantization levels k . One common approach is *resampling*, where the original time-series is either downsampled to reduce dimensionality or upsampled to increase granularity. Additionally, the choice of k directly determines the size of the resulting MTF matrix, as the output image has dimensions $k \times k$. Therefore, selecting an appropriate k allows the image resolution to be tailored to match the input requirements of downstream CNN architectures.

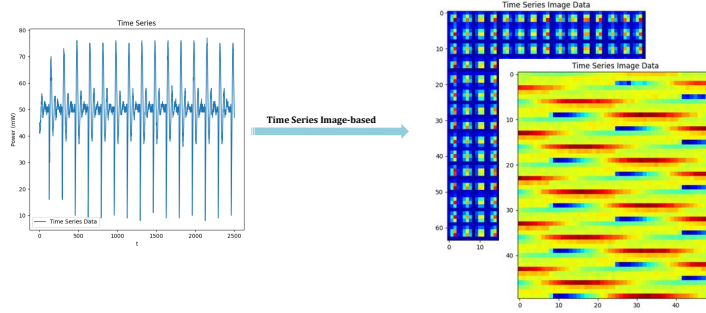


Fig. 1. Visualizing Time Series with MTF and Reshaping: (Left) Original Time Series, (Right) MTF and reshaping Representation.

Figure 1 shows the result of transforming a sample time-series signal using the reshaping and MTF method into an image-like matrix suitable for use by CNNs.

3.2. Dual-Path CNN Architecture for Trojan Detection and Classification

The image representations obtained through the transformations described in the previous sections are processed by two parallel CNN pathways within the architecture, as illustrated in Figure 2. Each CNN block applies convolutional filters to extract local features, followed by batch normalization to stabilize and accelerate training. A ReLU activation function introduces nonlinearity, enabling the model to capture complex spatial patterns. After each convolutional stage, a pooling layer

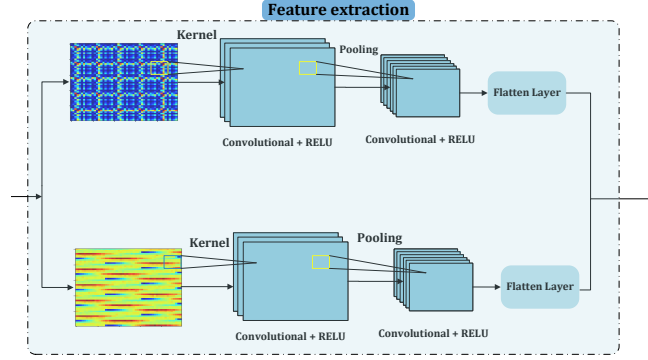


Fig. 2. The feature extraction block using two 2D image representations from time series data

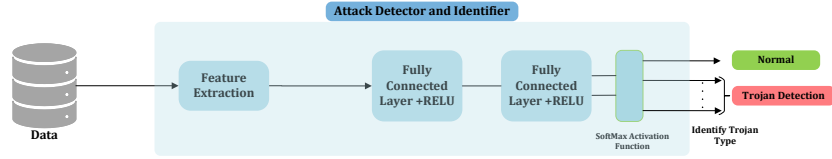


Fig. 3. One stage Attack Detector and Identifier.

reduces the spatial dimensions, preserving the most salient features while lowering computational cost. This process is repeated to ensure multi-scale feature extraction. The outputs from both CNN paths are then flattened, concatenated into a single feature vector, and passed through fully connected layers. Finally, a softmax classification layer outputs the predicted class, indicating whether the input corresponds to a normal instance or a specific type of Trojan activation.

The feature extraction module, which is integrated into the classifier component, forms a unified architecture for the simultaneous detection and identification of HT, as shown in Figure 3. In contrast to multi-stage frameworks, the proposed model follows a single-stage approach that directly classifies the input side-channel data into one of several predefined classes, including both normal and Trojan-infected instances. By using dual-path feature representations and fully linked decision layers, the model jointly determines whether a Hardware Trojan is present and, if so, identifies the specific type of attack. This streamlined architecture improves inference efficiency and reduces system complexity while maintaining high detection and classification accuracy, making it well suited for practical deployment in security-sensitive embedded systems.

3.3. Dataset Description and Experimental Setup

The dataset used in this paper is obtained from IEEE DataPort and originates from the Trust-Hub repository³⁹. It contains side-channel power traces recorded from an AES-128 cryptographic core performing 10,000 encryption operations per HT variant. Each sample is a univariate time-series of 2,500 points, representing the device’s power consumption during encryption. Data collection is conducted using Sakura-G boards, which provide high-resolution measurements under both normal and Trojan-triggered conditions. This length was selected to reliably capture subtle variations associated with Trojan activation. While the present experiments assume clean, high-resolution traces, future work will investigate the model’s robustness under noisy conditions and adversarially manipulated power measurements³⁹.

This work focuses on five HT variants: AES-T500, AES-T600, AES-T700, AES-T800, and AES-T1600. For each variant, power traces are collected under two operational modes: *Trojan Disabled*, where the Trojan circuitry is present but inactive, and *Trojan Triggered*, where the malicious logic is activated during execution.

All time-series signals undergo z-score normalization to reduce variation in signal amplitude and improve model convergence. The dataset is randomly partitioned into training and validation sets using an 80/20 split. Each trace is then transformed into two image-like representations using MTF and reshaping techniques. MTF images are generated using $k = 64$ quantization bins with a uniform strategy, producing 64×64 pixel matrices.

4. Results and Discussion

This section presents a comprehensive evaluation of the proposed time-series imaging-based framework for HT detection. The primary objective is to assess the classification accuracy, robustness, and generalization capability of the dual-path CNN architecture, which processes side-channel signals transformed into image representations using both MTF and reshaping techniques. Extensive experiments are conducted on the AES dataset to evaluate the model’s performance across multiple Trojan variants.

To ensure fair evaluation, 80% of the available signals are used for training and the remaining 20% for validation. The proposed model, implemented in PyTorch version 2.1.2 and illustrated in Figure 3, is tested under attack detection and attack classification settings. The results are organized to highlight baseline detection performance, class-specific evaluation using confusion matrices and ROC and PR metrics, architectural optimization through hyperparameter tuning, and latent space analysis using t-SNE for visualizing class separability and misclassification patterns.

4.1. Evaluation Performance of the Proposed Detector

The proposed dual-path model comprises two parallel branches for feature extraction: a CNN-based path and an MTF-based path. Each branch includes two con-

volutional layers, followed by batch normalization and max pooling operations, enabling progressive abstraction of spatial features while reducing dimensionality. In both branches, the convolutional layers use 32 and 64 filters, respectively, with a kernel size of 5×5 , stride of 1, and padding of 2. After the final pooling operation, each branch concludes with a fully connected layer that maps the extracted features into a 128-dimensional latent space. The feature vectors from the CNN and MTF branches are concatenated and passed to a shared fully connected layer with 128 units, followed by a final output layer with 6 nodes to perform multi-class classification of *Normal* and Trojan-infected samples. Table 1 summarizes the architecture parameters.

It should be noted that if the goal is to detect only a single type of Trojan (e.g., AES-T600), the same dual-path architecture can be employed with a binary output layer (2 nodes) instead of 6. In this binary configuration, the model is trained to distinguish between *Normal* samples and one specific Trojan class.

Table 1. Dual-Path CNN Architecture for AES Trojan Detection (CNN + MTF)

Branch	Layer Type	Filters/Units	Kernel Size	Stride	Padding
CNN	Convolutional Layer 1	32	5×5	1	2
	Batch Normalization	32	-	-	-
	Max Pooling	-	2×2	2	0
	Convolutional Layer 2	64	5×5	1	2
	Batch Normalization	64	-	-	-
	Max Pooling	-	2×2	2	0
	Fully Connected	128	-	-	-
MTF	Convolutional Layer 1	32	5×5	1	2
	Batch Normalization	32	-	-	-
	Max Pooling	-	2×2	2	0
	Convolutional Layer 2	64	5×5	1	2
	Batch Normalization	64	-	-	-
	Max Pooling	-	2×2	2	0
	Fully Connected	128	-	-	-
Combined	Concatenation (CNN + MTF)	256	-	-	-
	Fully Connected	128	-	-	-
	Output Layer (Softmax)	6 (Classes)	-	-	-

In this paper, we use the cross-entropy loss function for both binary and multi-class classification tasks. The binary classification loss is expressed as follows:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)] \quad (5)$$

where y represents the true label, $\hat{y}$ the predicted probability of the positive class, and N the number of samples in the dataset. Where, y_i indicates the presence or absence of an HT, and $\hat{y}_i$ is the predicted probability that the i -th sample contains an HT.

For the problem involving multi-class classification, the loss function is adapted

to multiple classes and is defined as follows ⁴⁰:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C [y_{i,c} \cdot \log(\hat{y}_{i,c})] \quad (6)$$

C represents the number of classes, $y_{i,c}$ indicates whether the class label c is the correct classification for the i -th observation, and $\hat{y}_{i,c}$ denotes the predicted probability that observation i belongs to class c . Accuracy for both binary and multi-class scenarios is calculated by ⁴⁰:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i = y_i) \quad (7)$$

where $\mathbb{1}(\cdot)$ is the indicator function that is 1 when $\hat{y}_i$ equals y_i and 0 otherwise.

To evaluate the effectiveness of the proposed HT detection methods, a comprehensive comparison with several established models is performed. These include the One-Class Support Vector Machine (OC-SVM), which is well known for its capability in anomaly detection, as well as three neural network architectures—MLP, CNN, and Hierarchical Temporal Memory (HTM). The HTM model is inspired by the human neocortex and is designed to handle temporal data. The performance metrics for these baseline models are based on the results reported in ³⁹. In addition, the comparison includes the Attention-Res-Attention (ARA) network proposed in ²⁴, which combines the ResNeXt architecture with an attention mechanism to enhance the detection of critical HT-related patterns.

Our model is evaluated independently for each AES Trojan type in a binary classification (normal vs. specific Trojan). Table 2 summarizes the achieved classification accuracy for each AES dataset. The proposed model achieves an accuracy of 98.5% for AES-T500, 93.25% for AES-T600, 100% for AES-T700 and AES-T800 and 75.40% for AES-T1600. These results show significant improvements over existing methods, especially for the more demanding AES-T600 and AES-T1600 benchmarks. It should be noted that for each binary classification experiment, the validation data is filtered to include only the *Normal* class and the specific Trojan class being evaluated. The model is then evaluated against this balanced subset, and the class-specific accuracy is calculated as the proportion of correctly predicted samples within this subset. This evaluation strategy ensures fair benchmarking and explicitly demonstrates the model's ability to distinguish each Trojan variant from *Normal* samples.

In addition to the binary evaluations, we have trained a unified multi-class model that is able to recognize and identify all five Trojan types together with the *normal* operation simultaneously. The architecture shown in Figure 3 is used with a six-node output layer, where each node corresponds to one of the six classes: *Normal*, AES-T500, AES-T600, AES-T700, AES-T800, and AES-T1600. The performance of this multi-class model is illustrated in Figure 5, which shows the training dynamics in terms of classification accuracy and loss across epochs. The results of the class-wise

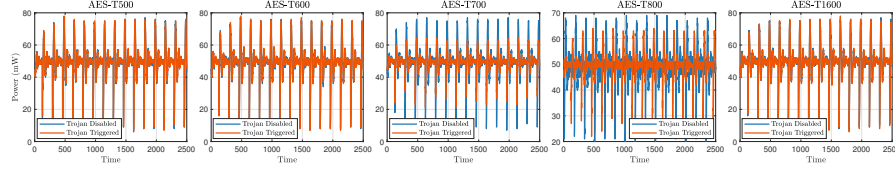


Fig. 4. Comparison of Trojan Disabled vs. Trojan Triggered for various datasets.

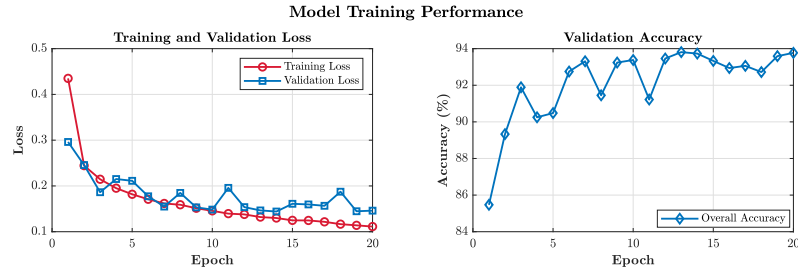


Fig. 5. Overall accuracy of the model across training epochs

prediction are summarized in Table 2. It should be noted that the baseline results for CNN, MLP, OC-SVM, HTM, and ARA are directly taken from the respective peer-reviewed publications cited in this paper. All of these studies evaluated their methods on the same AES Hardware Trojan dataset from TrustHub/IEEE DataPort used in our experiments, ensuring dataset comparability. We report the results exactly as published by the original authors to preserve methodological fidelity, acknowledging that preprocessing steps and hyperparameter choices follow those works and may differ from our proposed framework.

The next section presents a detailed metric-based analysis using the confusion matrix, as well as ROC and PR curves.

Table 2. Comparison of accuracy of Attack Detector with different methods on each AES Datasets

	AES-T500	AES-T600	AES-T700	AES-T800	AES-T1600
CNN	84.50%	55.50%	100%	100%	64.00%
MLP	84.00%	65.50%	100%	100%	70.50%
OC-SVM	89.70%	49.70%	90.20%	92.70%	50.20%
HTM ³⁹	100%	63.50%	98.10%	100%	59.60%
ARA ²⁴	100%	67.30%	100%	100%	63.10%
Our Method	98.50%	93.25%	100%	100%	75.40%

To assess the statistical robustness of the proposed dual-path CNN-MTF model, we performed stratified K-fold cross-validation (K=5), ensuring identical preprocessing and hyperparameters in each fold. In each run, the model was trained on 80% of the data and validated on the remaining 20%, with folds balanced by class to

prevent bias. The K-fold approach averages performance across all data partitions. This yields slightly lower average accuracies compared to the fixed split, particularly for more challenging classes (e.g., AES-T1600), as performance in harder folds impacts the average. Conversely, for classes such as AES-T700 and AES-T800, the model achieves perfect accuracy in most or all folds, with AES-T800 reaching 100% across all folds and AES-T700 exhibiting near-perfect performance. This consistent accuracy indicates the high separability of these classes. The per-class mean and standard deviation results across the folds are reported in Table 3. These results confirm that the model maintains high accuracy for all Trojan classes, with perfect classification for AES-T700 and AES-T800, and robust detection for more challenging cases such as AES-T1600.

Table 3. K-fold cross-validation accuracy (mean $\pm$ std) for each AES dataset using the proposed dual-path CNN-MTF model

	AES-T500	AES-T600	AES-T700	AES-T800	AES-T1600
Our Method (K-fold)	97.37% $\pm$ 0.16	92.37% $\pm$ 1.49	99.85% $\pm$ 0.30	100.0% $\pm$ 0.0	74.21% $\pm$ 1.22

4.2. *Class-wise Evaluation Using Confusion Matrix, ROC, and PR Metrics*

To complement the aggregate performance results discussed in the previous section, a detailed class-wise evaluation is conducted using the confusion matrix along with ROC and PR curves, computed on the held-out validation set.

While overall accuracy provides a useful summary, it may obscure performance disparities across individual classes—especially in multi-class or imbalanced settings. The confusion matrix offers a detailed overview of classification behavior by displaying the counts of true positives, false positives, true negatives, and false negatives for each class⁴¹. This enables the identification of specific sources of error, such as systematic misclassification between similar Trojan types or mislabeling of benign samples. ROC curves characterize the trade-off between true positive rate (sensitivity) and false positive rate across different threshold values, making them well-suited for evaluating classifier discriminability⁴¹. However, under class imbalance, PR curves often provide more informative insights, as they focus on the trade-off between precision and recall, and are less affected by the prevalence of true negatives⁴². For this reason, both metrics are used together to evaluate the robustness and reliability of the proposed model in detecting subtle or low-signal Trojan activations.

Figure 6 shows the 6×6 normalized confusion matrix, which includes the class *Normal* and the five Trojan classes: AES-T500, AES-T600, AES-T700, AES-T800, and AES-T1600. The matrix exhibits a strong diagonal dominance, indicating a high overall classification accuracy. Remarkably, the model achieves perfect classi-

fication for AES-T800 and near-perfect classification for AES-T700, with only a single sample misclassified in the latter case. Minor misclassifications are observed between AES-T600 and AES-T500 as well as between AES-T1600 and the *Normal* class. Nevertheless, the false positive rate for the *Normal* class remains below 2%, highlighting the ability of the model to minimize false positives despite the presence of different and possibly overlapping Trojan patterns.

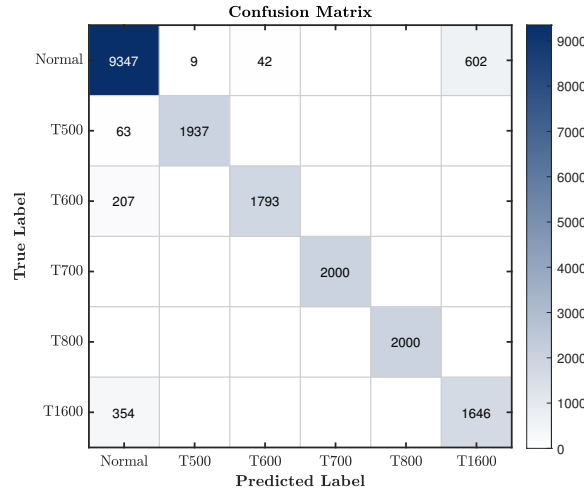


Fig. 6. Normalised confusion matrix of the proposed model on the validation set.

Figures 7 and 8 illustrate the one-vs-rest ROC and PR curves for each class compared to the rest. These plots provide a detailed overview of the classifier’s performance across different decision thresholds and emphasize its robustness in distinguishing the *Normal* class from the five Trojan classes. The ROC curves in Figure 7 show high true positive rates (TPRs) with very low false positive rates (FPRs) across all classes. The curves rise steeply and reach saturation near the upper-left corner, indicating strong separability between classes. In particular, the curves for AES-T500, AES-T700, and AES-T800 approach the ideal boundary, reflecting excellent detection performance. The *Normal* class also exhibits favorable ROC characteristics, with minimal overlap with Trojan distributions.

Figure 8 shows the corresponding PR curves, which further confirm the effectiveness of the classifier under class imbalance. Most classes achieve near-perfect precision and recall across a wide range of thresholds. Notably, the AES-T1600 class shows a gradual decline in precision at lower recall values, consistent with its relatively lower F1-score, while still achieving a strong area under the curve. Overall, the classifier demonstrates consistently high discriminatory power, with all AUC-ROC values above 0.98 and AUC-PR values above 0.81, as summarized in

Table 4. These results validate the proposed model’s reliability, even in scenarios characterized by class imbalance and overlapping feature distributions.

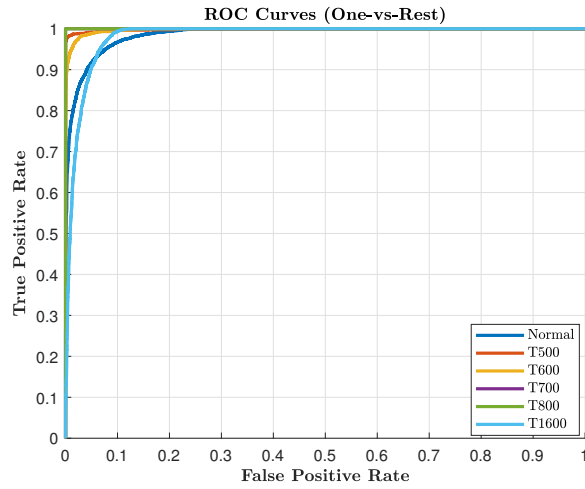


Fig. 7. One-vs-rest ROC curves for each class.

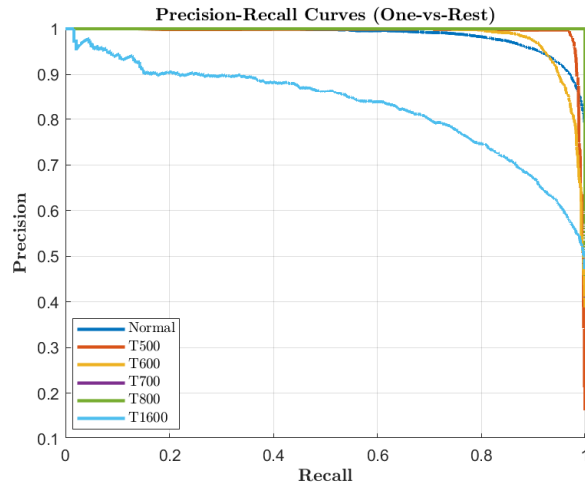


Fig. 8. One-vs-rest precision-recall curves for each class.

Table 4 summarizes the per-class validation metrics corresponding to the final confusion matrix shown in Figure 6. The proposed model achieves macro-averaged

precision, recall, and F1-score values of 0.93, 0.95, and 0.94, respectively, where macro-averaging treats all classes equally by computing the unweighted mean of the metric across classes. This indicates balanced performance across both Trojan and *Normal* classes. The AES-T700 and AES-T800 classes are perfectly detected, with F1-scores of 1.00, indicating the model’s strong reliability in identifying frequently triggered Trojan types. The *Normal* class also exhibits excellent generalization, achieving a recall of 0.98, which ensures that benign instances are rarely misclassified as malicious. Among the Trojan classes, AES-T1600 presents the greatest challenge, with an F1-score of 0.77 and an AUC-PR of 0.828. Its reduced precision is primarily attributed to feature overlap with the *Normal* class, as reflected by the relatively high misclassification rate. Similarly, AES-T600 exhibits moderate degradation, with an F1-score of 0.92, suggesting occasional confusion with both AES-T500 and *Normal* samples. Despite these challenges, all class-specific metrics exceed baseline performance levels reported in previous studies, thereby validating the discriminative capability of the proposed dual-path CNN architecture. These results also underscore the benefit of transforming side-channel traces into structured image representations, reinforcing the model’s suitability for deployment in practical Hardware Trojan detection systems.

Table 4. Per-class validation metrics of the proposed model.

Class	Precision	Recall	F1-Score	AUC-ROC	AUC-PR
Normal	0.950	0.980	0.960	0.987	0.981
AES-T500	0.990	0.970	0.980	0.998	0.993
AES-T600	0.940	0.900	0.920	0.996	0.983
AES-T700	1.000	1.000	1.000	1.000	1.000
AES-T800	1.000	1.000	1.000	1.000	1.000
AES-T1600	0.730	0.820	0.770	0.980	0.828
Macro Avg.	0.930	0.950	0.940	0.993	0.964

Figure 9 provides a visual overview of the class-wise Precision, Recall, and F1-score values, complementing the detailed results in Table 4.

4.3. Hyperparameter Optimization of CNN and MTF Paths

To improve the performance of the proposed dual-path CNN-MTF architecture, a comprehensive hyperparameter optimization procedure is conducted independently for both the CNN and MTF branches. The objective is to identify the most effective architectural parameters—specifically, the number of convolutional filters and kernel dimensions—that maximize validation accuracy while minimizing training loss.

The paper systematically evaluates several configurations by varying combinations of convolutional filter sizes and kernel dimensions. For both the CNN and

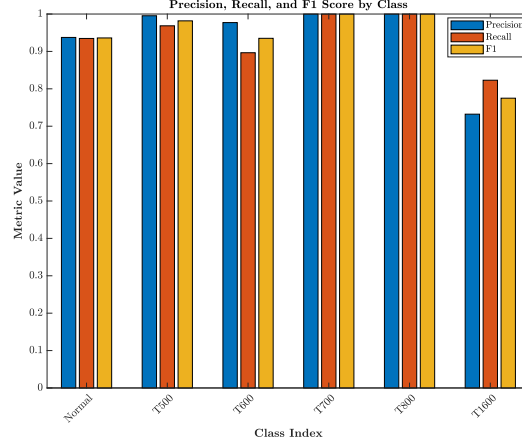
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Fig. 9. Bar plot of per-class Precision, Recall, and F1-Score. This figure visually complements Table 4.

MTF branches, three filter settings—(32, 64), (16, 32), and (128, 64)—are explored to assess the impact of network depth and representational capacity. Kernel sizes of 3×3 and 5×5 are examined to evaluate their effect on spatial feature extraction. To maintain consistency across all experiments, the dropout rate is fixed at 0.15, providing uniform regularization. Additionally, the batch size and learning rate are kept constant at 64 and 0.0001, respectively, based on preliminary empirical evaluations. The model variants evaluated during this paper are labeled as A_1 through A_4 , as summarized in Table 5. Each configuration corresponds to a unique combination of CNN filters, MTF filters, and kernel sizes. This labeling scheme facilitates a direct comparison of the effect of architectural choices on classification accuracy.

Table 5. Summary of Model Configurations and Accuracy

Model	CNN Filters	MTF Filters	Kernels (CNN / MTF)	Best Accuracy (%)
A_1	(128, 64)	(16, 32)	$5 \times 5 / 5 \times 5$	94.22
A_2	(16, 32)	(128, 64)	$5 \times 5 / 5 \times 5$	93.08
A_3	(16, 32)	(128, 64)	$3 \times 3 / 5 \times 5$	92.77
A_4	(16, 32)	(32, 64)	$3 \times 3 / 5 \times 5$	92.77

Figure 10 presents the training loss and validation accuracy curves for the best-performing configuration. The model exhibits stable convergence, with minimal variance between training and validation trajectories, indicating strong generalization capability and robustness against overfitting. This optimal configuration employs 128 and 64 filters in the CNN branch, 16 and 32 filters in the MTF branch, and kernel sizes of 5×5 in both pathways.

The comparative analysis across all configurations in Table 5 clearly identifies Model A_1 as the most effective. In addition to achieving the highest validation accuracy of 94.22%, it also maintains one of the lowest validation losses (0.1359), reflecting both learning efficiency and predictive consistency. This architecture achieves a favorable balance between deep spatial representation in the CNN path and lightweight temporal modeling in the MTF path.

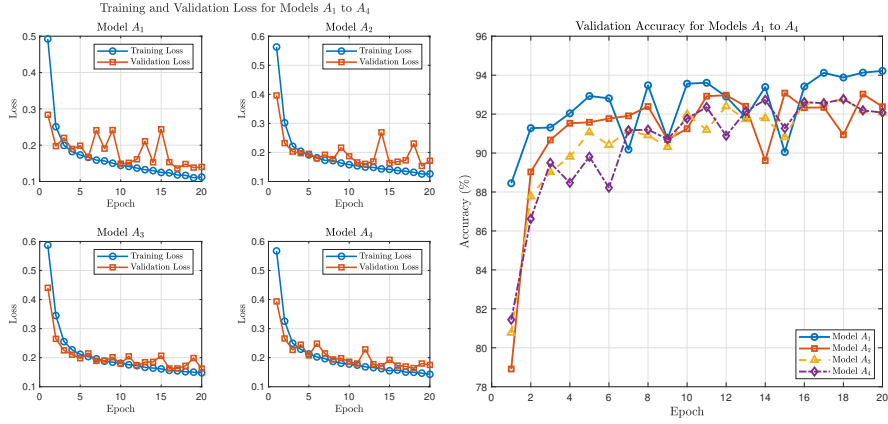


Fig. 10. Training loss (left) and validation accuracy (right) across epochs for the best hyperparameter configuration.

4.4. Latent Feature Space Visualization and Misclassification Analysis

To gain deeper insight into the internal behavior of the proposed dual-path CNN architecture, we employ a latent-space visualization technique based on the t-SNE algorithm. As a nonlinear dimensionality reduction method, t-SNE is widely used to explore the structure of high-dimensional feature representations in deep learning models by projecting them into a low-dimensional space that preserves local neighborhood relationships⁴³. This allows for intuitive, visual assessment of class separability and the identification of ambiguous or overlapping feature regions that may lead to misclassification.

In this paper, t-SNE is applied to the final 128-dimensional latent features output by the shared fully connected layer preceding the classifier. Figure 11 displays the resulting projection for a representative subset of validation samples, with each point color-coded according to its ground-truth class label. The projected clusters reveal clear semantic separation among most Trojan types, particularly AES-T700 and AES-T800, whose latent representations form compact, non-overlapping clusters. The *Normal* class also exhibits substantial structure, forming multiple contiguous subgroups that likely correspond to intraclass variability due to noise or

environmental conditions. Notably, AES-T600 and AES-T1600 samples show partial overlap with both the *Normal* and AES-T500 classes.

To further analyze model performance, Figure 12 overlays the misclassified samples on the same t-SNE projection. Misclassified points are marked as red crosses, while correctly classified ones appear as green circles. These errors tend to cluster in or near the ambiguous boundaries previously identified. In particular, most false positives—*Normal* samples incorrectly classified as Trojan—occur near the boundary shared with AES-T500 and AES-T600. Conversely, several AES-T1600 instances are misclassified as *Normal*, highlighting the model’s difficulty in detecting weak or stealthy Trojan activity. This analysis suggests that the dual-path model is effective at learning discriminative features for clearly manifested Trojans (e.g., AES-T700, AES-T800), but struggles with edge cases involving subtle perturbations or low-SNR scenarios. Potential future improvements include incorporating attention mechanisms, deeper convolutional hierarchies, or auxiliary loss functions that penalize latent-space overlap. Training strategies such as hard-negative mining or contrastive learning may also enhance the separation of ambiguous classes.

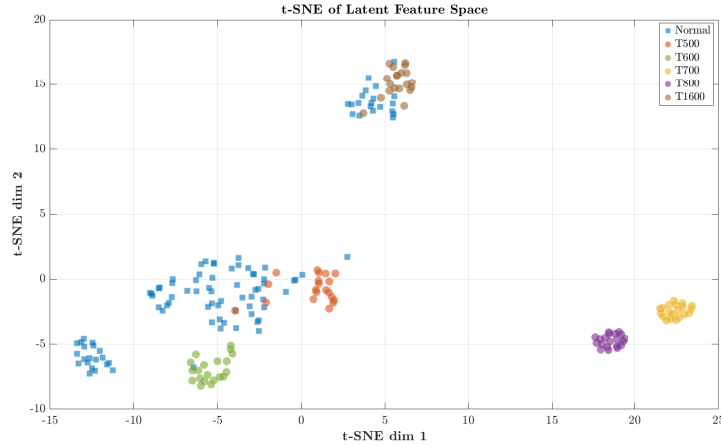


Fig. 11. t-SNE projection of latent feature space colored by ground-truth class. Compact clusters are observed for T700 and T800, while overlapping regions are evident between *Normal*, T500, and T600.

4.5. Computational Cost Analysis

To assess the practicality of deploying the proposed dual-path CNN model in resource-constrained environments, we evaluated its computational complexity, model size, and latency. The model contains approximately 3.42 million trainable parameters, with a total size of 13.03 MB when stored in 32-bit floating point

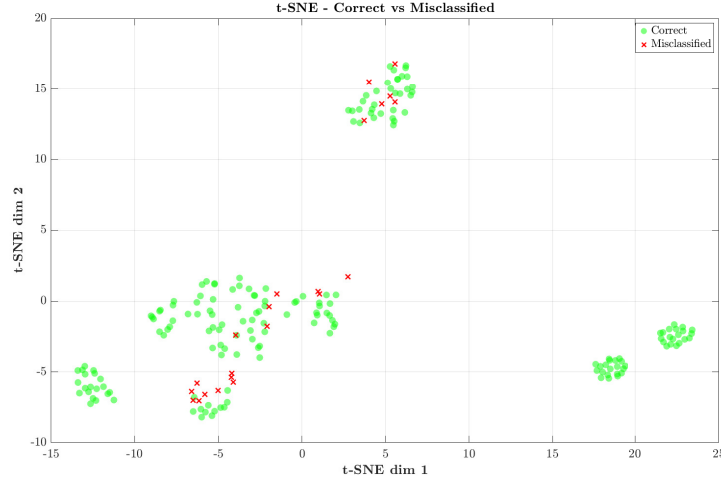


Fig. 12. t-SNE visualization showing correctly classified (green circles) and misclassified (red crosses) validation samples. Misclassifications are concentrated in ambiguous feature regions.

format. The number of floating point operations (FLOPs) per inference pass is estimated at 93.65 million. Inference latency was measured at approximately 0.817 milliseconds per input sample on an NVIDIA Quadro RTX 6000 GPU, indicating that the model is capable of real-time or near-real-time performance. Memory usage during a forward and backward pass is approximately 18.76 MB, including activation storage and parameter buffers. These results confirm that the model is computationally lightweight and suitable for embedded or real-time HT detection applications.

4.6. Limitation & Future Work.

While the proposed model demonstrates strong performance on the clean, labeled traces from TrustHub, we acknowledge that real-world deployment may involve additional complexities. In particular, the separability between Trojan-triggered and Trojan-disabled traces may be affected by process variations, environmental noise, or side-channel interference. The dataset used in this work does not include noise-augmented samples or out-of-distribution (OOD) Trojan variants. As such, the reported performance assumes clean separability under controlled conditions. Future work will focus on evaluating robustness under synthetic noise injection and testing generalization to OOD inputs, such as novel Trojan types or altered hardware configurations.

5. Conclusion

This paper proposed a novel dual-path convolutional neural network (CNN) framework for detecting and identifying Hardware Trojans (HTs) in integrated circuits using side-channel analysis. By transforming power traces into image-like representations through both Markov Transition Field (MTF) encoding and time-series reshaping, the model effectively uses 2D CNN architectures to extract complementary spatial features from side-channel signals. The dual-path architecture enhances the discriminative capacity of the network by integrating distinct feature perspectives from parallel branches. Extensive experiments conducted on the Advanced Encryption Standard (AES) datasets from TrustHub and IEEE DataPort demonstrated that the proposed method significantly outperforms several existing baselines, including OC-SVM, MLP, CNN, HTM, and attention-based networks. In binary classification settings, the model achieved accuracies of 98.5% for AES-T500, 93.25% for AES-T600, and 75.40% for AES-T1600, highlighting its effectiveness even against weakly activated Trojans. A multiclass evaluation further validated its generalization ability across all six classes, including the *Normal* case and five distinct Trojan variants. A comprehensive class-wise analysis using confusion matrices, ROC, and PR curves confirmed the model's robustness, with macro-averaged precision, recall, and F1-scores reaching 0.93, 0.95, and 0.94, respectively. Latent feature space visualization using t-SNE revealed well-separated clusters for most classes, while also highlighting regions of ambiguity for low-SNR Trojans like AES-T1600. Misclassification analysis provided insights into potential areas for model improvement. Hyperparameter optimization experiments explored the influence of filter sizes and kernel dimensions, ultimately identifying the most effective configuration, which balances deep CNN spatial encoding with lightweight MTF-based temporal modeling. Future work will focus on enhancing the model's resilience to noise and its ability to generalize across additional datasets and Trojan types. Potential improvements include integrating attention mechanisms, contrastive training objectives, and advanced regularization strategies to further reduce misclassification in ambiguous scenarios.

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References

1. R. S. Chakraborty, S. Narasimhan and S. Bhunia, Hardware Trojan: Threats and emerging solutions, in *2009 IEEE International High Level Design Validation and Test Workshop*, (IEEE, nov 2009), pp. 166–171.

2. S. Sun, H. Zhang, X. Cui, L. Dong and X. Fang, Electromagnetic Side-Channel Hardware Trojan Detection Based on Transfer Learning, *IEEE Transactions on Circuits and Systems II: Express Briefs* **69**(3) (2022) 1742–1746.
3. A. Amelian and S. E. Borujeni, A Side-Channel Analysis for Hardware Trojan Detection Based on Path Delay Measurement, *Journal of Circuits, Systems and Computers* **27**(aug 2018) p. 1850138.
4. J. Francq and F. Frick, Introduction to Hardware Trojan Detection Methods, in *Design, Automation & Test in Europe Conference & Exhibition (DATE), 2015*, (IEEE Conference Publications, New Jersey, 2015), pp. 770–775.
5. A. Golabi, A. Erradi and A. Tantawy, Towards automated hazard analysis for CPS security with application to CSTR system, *Journal of Process Control* **115**(jul 2022) 100–111.
6. M. Xue, C. Gu, W. Liu, S. Yu and M. O'Neill, Ten years of hardware Trojans: a survey from the attacker's perspective, *IET Computers & Digital Techniques* **14**(nov 2020) 231–246.
7. Y. Liu, Y. Zhao, J. He and R. Xin, A Statistical Test Generation Based on Mutation Analysis for Improving the Hardware Trojan Detection, *Journal of Circuits, Systems and Computers* **29**(mar 2020) p. 2050049.
8. E. Vaddi, K. Gaddam, R. K. Maniam, S. A. Mallavajjala, S. Dasari and N. D. M., Detection and diagnosis of hardware trojan using power analysis, in *Security in Computing and Communications*, eds. J. H. Abawajy, S. Mukherjea, S. M. Thampi and A. Ruiz-Martínez (Springer International Publishing, Cham, 2015), pp. 519–529.
9. Z. Pan and P. Mishra, A Survey on Hardware Vulnerability Analysis Using Machine Learning, *IEEE Access* **10** (2022) 49508–49527.
10. C. Dong, Y. Liu, J. Chen, X. Liu, W. Guo and Y. Chen, An unsupervised detection approach for hardware trojans, *IEEE Access* **8**(Ic) (2020) 158169–158183.
11. A. Golabi, A. Erradi, A. Bensaid, A. Al-Ali and U. Qidwai, A dual-stage deep learning approach for robust detection and identification of hardware trojans using monte-carlo dropout, *International Journal of Information Security* **24**(3) (2025) 1–20.
12. A. Mondal, A. Ghosh, S. Karmakar, M. H. Mahalat, S. Roy and B. Sen, Identification of Hardware Trojan in Gate-Level Netlist, *Journal of Circuits, Systems and Computers* **33**(jun 2024).
13. T.-T. Hoang, T.-H. Tran, V.-P. Hoang, X.-N. Tran and C.-K. Pham, Hardware trojan detection techniques using side-channel analysis, in *2019 6th NAFOSTED Conference on Information and Computer Science (NICS)*, 2019, pp. 528–533.
14. F. K. Lodhi, S. R. Hasan, O. Hasan and F. Awwadl, Power profiling of microcontroller's instruction set for runtime hardware Trojans detection without golden circuit models, *Proceedings of the 2017 Design, Automation and Test in Europe, DATE 2017* (2017) 294–297.
15. Z. Huang, C. Xie, Z. Li, M. Du and Q. Wang, A Hardware Trojan Detection and Diagnosis Method for Gate-Level Netlists Based on Different Machine Learning Algorithms, *Journal of Circuits, Systems and Computers* **31**(may 2022).
16. P. Ma, Z. Wang and Y. Wang, A Pre-Silicon Detection Based on Deep Learning Model for Hardware Trojans, *Journal of Circuits, Systems and Computers* **33**(may 2024).
17. H. Mohammed, T. A. Odetola, S. R. Hasan, S. Stissi, I. Garlin and F. Awwad, (HIA-DIoT): Hardware Intrinsic Attack Detection in Internet of Things; Leveraging Power Profiling, *Midwest Symposium on Circuits and Systems* **2019-August** (2019) 852–855.
18. T. Dhar, R. Das, C. Giri and S. K. Roy, Threshold Analysis Using Probabilistic Xgboost Classifier for Hardware Trojan Detection, *Journal of Electronic Testing: Theory and Applications (JETTA)* **39**(4) (2023) 447–463.

24 Arash Golabi et al.

19. N. P. Bhatta and F. Amsaad, MI assisted techniques in power side channel analysis for trojan classification, *Cluster Computing* **28**(jun 2025) p. 157.
20. V. S. Rathor, S. Podder and S. Dubey, An ensemble learning model for hardware Trojan detection in integrated circuit design, *Computers and Electrical Engineering* **123**(PB) (2025) p. 110090.
21. A. Sagu, N. S. Gill and P. Gulia, Hybrid Deep Neural Network Model for Detection of Security Attacks in IoT Enabled Environment, *International Journal of Advanced Computer Science and Applications* **13**(1) (2022) 120–127.
22. M. Rahimifar, H. Jahanirad and M. Fathi, Deep transfer learning approach for digital circuits vulnerability analysis, *Expert Systems with Applications* **237**(mar 2024) p. 121757.
23. W. Tang, J. Su, J. He and Y. Gao, A Deep Learning Method Based on the Attention Mechanism for Hardware Trojan Detection, *Electronics (Switzerland)* **11**(15) (2022).
24. S. Chen, T. Wang, Z. Huang and X. Hou, Detection method of Golden Chip-Free Hardware Trojan based on the combination of ResNeXt structure and attention mechanism, *Computers and Security* **134**(0123456789) (2023) p. 103428.
25. B. Dakhale, K. Vipinkumar, K. Narotham, S. Kadam, A. A. Bhurane and A. G. Kothari, Automated detection of hardware trojans using power side-channel analysis and vgg-net, in *2023 2nd International Conference on Paradigm Shifts in Communications Embedded Systems, Machine Learning and Signal Processing (PCEMS)*, 2023, pp. 1–5.
26. A. Golabi, A. Erradi, A. Bensaid, A. Al-Ali and U. Qidwai, Enhancing hardware trojan detection: A dual-path cnn approach to side-channel analysis, in *2024 International Conference on Microelectronics (ICM)*, 2024, pp. 1–6.
27. Z. Wang and T. Oates, Encoding time series as images for visual inspection and classification using tiled convolutional neural networks, *AAAI Workshop - Technical Report WS-15-14* (2015) 40–46.
28. D. Dias, A. Pinto, U. Dias, R. Lamparelli, G. Le Maire and R. D. S. Torres, A Multirepresentational Fusion of Time Series for Pixelwise Classification, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* **13**(March) (2020) 4399–4409.
29. G. R. Garcia, G. Michau, M. Ducoffe, J. S. Gupta and O. Fink, Temporal signals to images: Monitoring the condition of industrial assets with deep learning image processing algorithms, *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability* **236**(4) (2022) 617–627.
30. N. Wang and H. Kang, An Enhanced Detection Method of Hardware Trojan Based on CNN- Attention-LSTM, *Computer Networks and Communications* **2**(2) (2024) 361–373.
31. S. Bhunia, M. S. Hsiao, M. Banga and S. Narasimhan, Hardware trojan attacks: Threat analysis and countermeasures, *Proceedings of the IEEE* **102**(8) (2014) 1229–1247.
32. A.-A. Semenoglou, E. Spiliotis and V. Assimakopoulos, Image-based time series forecasting: A deep convolutional neural network approach, *Neural Networks* **157**(jan 2023) 39–53.
33. A. Ben Said and A. Erradi, Deep-Gap: A Deep Learning Framework for Forecasting Crowdsourcing Supply-Demand Gap Based on Imaging Time Series and Residual Learning, in *2019 IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, (IEEE, dec 2019), pp. 279–286.
34. Q. Ge, C. Wei and D. Wu, Abnormal Network User Detection Method based on the Hybrid Neural Network from the Perspective of Multimodal Data Fusion, *Journal of*

- Circuits, Systems and Computers* **0**(mar 2025).
35. K. Srinivasan, J. Dauwels and M. R. Reddy, A two-dimensional approach for lossless EEG compression, *Biomedical Signal Processing and Control* **6**(oct 2011) 387–394.
 36. Y. Wu, B. Wang, R. Yuan and J. Watada, A Gramian angular field-based data-driven approach for multiregion and multisource renewable scenario generation, *Information Sciences* **619**(jan 2023) 578–602.
 37. Z. Wang and T. Oates, Imaging time-series to improve classification and imputation, *IJCAI International Joint Conference on Artificial Intelligence* **2015-January**(Ijcai) (2015) 3939–3945.
 38. A. Ben Said, A.-S. G. Abdel-Salam and K. A. Hazaa, Performance prediction in online academic course: a deep learning approach with time series imaging, *Multimedia Tools and Applications* **83**(nov 2023) 55427–55445.
 39. S. Faezi, R. Yasaei, A. Barua and M. A. A. Faruque, Brain-Inspired Golden Chip Free Hardware Trojan Detection, *IEEE Transactions on Information Forensics and Security* **16** (2021) 2697–2708.
 40. I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning* (MIT Press, 2016). <http://www.deeplearningbook.org>.
 41. T. Fawcett, An introduction to roc analysis, *Pattern Recognition Letters* **27**(8) (2006) 861–874.
 42. T. Saito and M. Rehmsmeier, The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets, *PloS One* **10**(3) (2015) p. e0118432.
 43. L. van der Maaten and G. Hinton, Visualizing data using t-sne, *Journal of Machine Learning Research* **9**(Nov) (2008) 2579–2605.